QUBO Formulation: Single-Track Railway Dispatching Problem

Single-Track Railway Dispatching Problem

Problem Statement. Given a set of N trains that must traverse a shared single-track segment, each train $i \in \{0, \ldots, N-1\}$ is offered a set of K_i mutually exclusive timing options. Each option (i, j), where $j \in \{0, \ldots, K_i - 1\}$, incurs a known delay cost $c_{i,j}$. Certain pairs of options $((i, j), (k, l))$ are incompatible due to track occupancy conflicts, denoted by the set $\mathcal{C}$. The task is to select exactly one timing option for every train such that no two selected options conflict, while minimizing the total dispatch delay cost.

Decision Variables. We introduce binary decision variables $x_{i,j} \in \{0, 1\}$ for each train $i \in \{0, \ldots, N-1\}$ and each option $j \in \{0, \ldots, K_i - 1\}$. The variable $x_{i,j}$ equals 1 if train i selects option j, and 0 otherwise. The full decision vector is $\mathbf{x} = \{x_{i,j}\}_{i,j}$.

QUBO Derivation. Step 1. The original optimization problem is a minimization of the total delay cost plus a penalty for selecting conflicting options:

$$\min_{\mathbf{x}} \quad H_{\text{obj}}(\mathbf{x}) = \sum_{i=0}^{N-1} \sum_{j=0}^{K_i-1} c_{i,j} x_{i,j} + M \sum_{((i,j),(k,l)) \in \mathcal{C}} x_{i,j} x_{k,l}, \tag{1}$$

where $M > 0$ is a sufficiently large penalty constant. Since QUBO solvers minimize a quadratic form, we directly construct the Hamiltonian $H(\mathbf{x})$ to match this objective.

Step 2. The problem is subject to two types of constraints. First, each train must select exactly one option:

$$\sum_{j=0}^{K_i-1} x_{i,j} = 1, \quad \forall i \in \{0, \ldots, N-1\}. \tag{2}$$

Second, no two conflicting options may be selected simultaneously:

$$x_{i,j} + x_{k,l} \leq 1, \quad \forall ((i, j), (k, l)) \in \mathcal{C}. \tag{3}$$

Step 3. We convert the equality constraints into quadratic penalty terms using a weight $\lambda > 0$. The penalty for train i is:

$$P_i(\mathbf{x}) = \lambda \left(\sum_{j=0}^{K_i-1} x_{i,j} - 1 \right)^2 . \tag{4}$$

Expanding the square yields:

$$P_i(\mathbf{x}) = \lambda \left(\sum_{j=0}^{K_i-1} x_{i,j}^2 - 2 \sum_{j=0}^{K_i-1} x_{i,j} + 1 + \sum_{j=0}^{K_i-1} \sum_{\substack{l=0 \\ l \neq j}}^{K_i-1} x_{i,j} x_{i,l} \right). \tag{5}$$

Using the idempotent property of binary variables, $x_{i,j}^2 = x_{i,j}$, and noting that the double sum counts each distinct pair twice, we obtain:

$$P_i(\mathbf{x}) = \lambda \left(\sum_{j=0}^{K_i-1} x_{i,j} - 2 \sum_{j=0}^{K_i-1} x_{i,j} + 1 + 2 \sum_{0 \leq j < l \leq K_i-1} x_{i,j} x_{i,l} \right). \tag{6}$$

Collecting linear terms gives the general penalty contribution for train i:

$$P_i(\mathbf{x}) = \lambda \left(- \sum_{j=0}^{K_i-1} x_{i,j} + 2 \sum_{0 \leq j < l \leq K_i-1} x_{i,j} x_{i,l} + 1 \right). \tag{7}$$

The conflict constraints are already enforced by the quadratic terms in $H_{\text{obj}}(\mathbf{x})$ with coefficient M, so no additional penalty expansion is required for them.

Step 4. Combining the objective and all equality penalties, the total QUBO Hamiltonian is:

$$H(\mathbf{x}) = H_{\text{obj}}(\mathbf{x}) + \sum_{i=0}^{N-1} P_i(\mathbf{x}). \tag{8}$$

Step 5. Reading off the coefficients from the expanded Hamiltonian, the QUBO matrix entries $Q_{(i,j),(k,l)}$ and constant offset c are given in closed form by:

$$Q_{(i,j),(i,j)} = c_{i,j} - \lambda, \tag{9}$$

$$Q_{(i,j),(k,l)} = M \quad \text{if } ((i,j),(k,l)) \in \mathcal{C}, \tag{10}$$

$$Q_{(i,j),(i,l)} = 2\lambda \quad \text{for } 0 \leq j < l \leq K_i - 1, \tag{11}$$

$$Q_{(i,j),(k,l)} = 0 \quad \text{otherwise}, \tag{12}$$

with the constant offset $c = N\lambda$. The QUBO is then expressed as $H(\mathbf{x}) = \mathbf{x}^\top Q \mathbf{x} + c$.

Penalty Weight Justification. The equality penalty weight λ must be chosen large enough to dominate any potential reduction in delay costs that could be achieved by violating the exactly-one constraint. Specifically, if a train selects zero or multiple options, the penalty incurred is at least λ. Therefore, selecting $\lambda > \max_{i,j} c_{i,j}$ guarantees that any infeasible assignment has a strictly higher energy than the best feasible assignment. The conflict penalty M must similarly satisfy $M > \sum_{i,j} c_{i,j}$ to ensure that violating a track conflict is never cost-effective.

Correctness Argument. Minimizing $H(\mathbf{x})$ over $\{0,1\}^n$ recovers the optimal dispatch schedule because the penalty terms are non-negative and vanish exactly on the feasible set. Any infeasible assignment violates at least one constraint, incurring a penalty strictly greater than the maximum possible objective value of any feasible assignment. Consequently, the global minimum of $H(\mathbf{x})$ must lie within the feasible region. Within this region, $H(\mathbf{x})$ reduces exactly to the original delay cost objective. Thus, the unconstrained QUBO minimum corresponds precisely to the constrained optimization optimum.

Illustrative Example. Consider a concrete instance with $N = 3$ trains and $K_i = 4$ options per train, yielding 12 binary variables indexed by $u = 4i + j \in \{0, \ldots, 11\}$. Let the delay costs be $c_{i,j}$, the equality penalty weight be $\lambda = 1000$, and the conflict penalty be M. The conflict set $\mathcal{C}$ contains three pairs: $((0,0),(1,1))$, $((0,1),(1,0))$, and $((1,0),(2,2))$, corresponding to variable indices $(0,5)$, $(1,8)$, and $(4,10)$.

Instantiating the general formulas, the Hamiltonian becomes:

$$H(\mathbf{x}) = \sum_{u=0}^{11}(c_{i,j} - 1000)x_u + 2000\sum_{i=0}^{2}\sum_{0\le j<l\le 3} x_{4i+j}x_{4i+l} + M(x_0x_5 + x_1x_8 + x_4x_{10}) + 3000. \tag{13}$$

The corresponding Q matrix entries are:

$$Q_{u,u} = c_{i,j} - 1000, \quad u = 4i + j, \tag{14}$$

$$Q_{0,5} = Q_{5,0} = M, \quad Q_{1,8} = Q_{8,1} = M, \quad Q_{4,10} = Q_{10,4} = M, \tag{15}$$

$$Q_{u,v} = 2000 \quad \text{for } u, v \text{ belonging to the same train with } j < l, \tag{16}$$

$$Q_{u,v} = 0 \quad \text{otherwise}, \tag{17}$$

and the constant offset is $c = 3000$. Solving this 12-variable QUBO yields the optimal binary vector $\mathbf{x}^*$, from which the selected timing option for each train is recovered by identifying the index j where $x^*_{i,j} = 1$. The minimum energy value equals the minimum feasible delay cost plus the constant offset.